

BAAZ JKBOSE



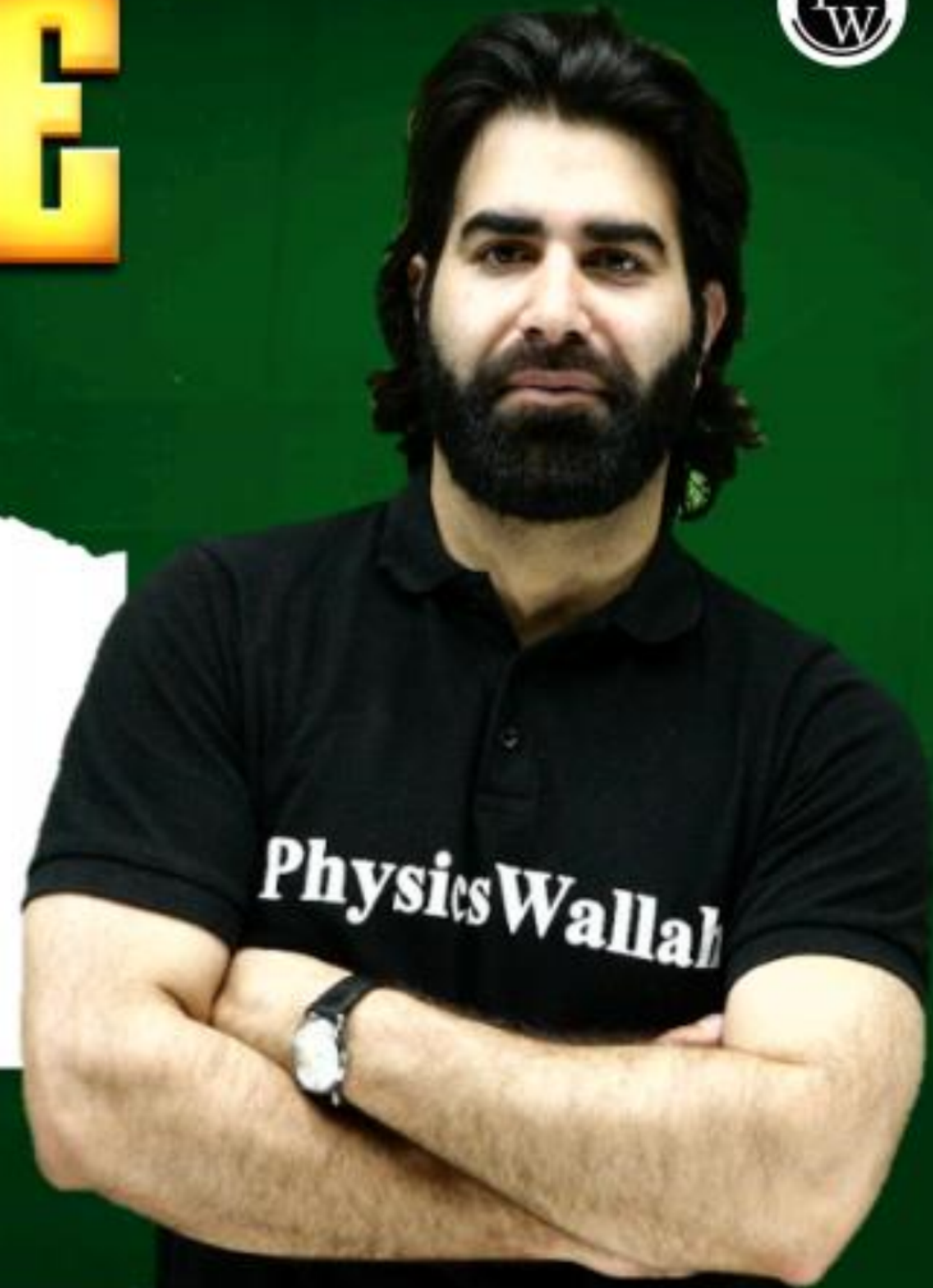
2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-02

Zoology

Dr Adil Altaf sir



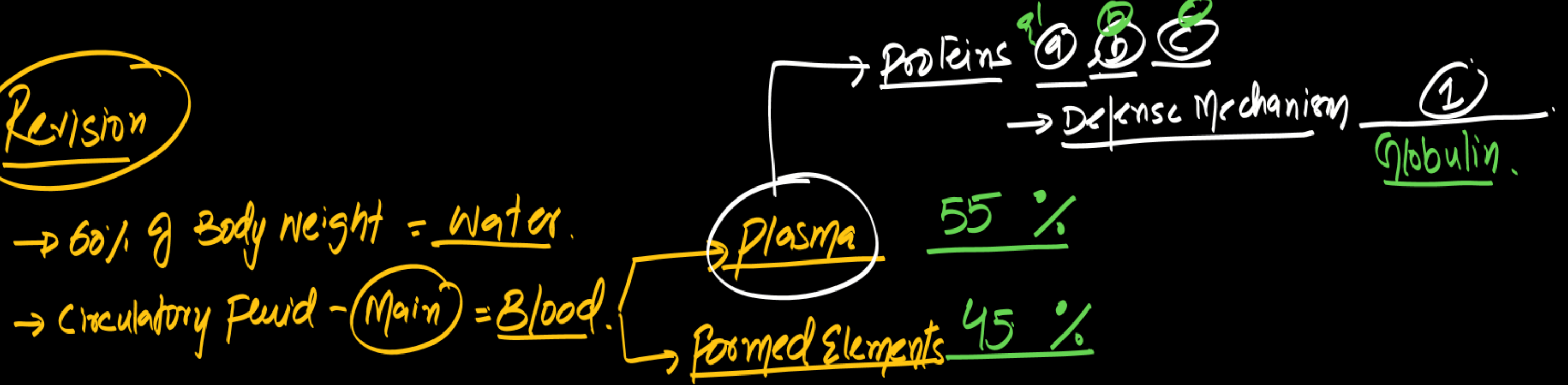
Topics To Be Covered

1 Blood Groups

Revision

→ 60% of Body weight = Water.

→ Circulatory Fluid - (Main) = Blood.



→ RBC = Human - Shape = Round, Biconcave



H/W - Hb Metabolism

- Bi product (Waste) =
(Liver Remove)

Billiverdin, Bilisubin
(green) (yellow)

→ Rise in Blood.
= Jaundice

Formed Elements.

① RBC - Discussed.

② WBC (White Blood Corpuscles)



- ↳ These are also called as → Leukocytes (Leucocytes)
- ↳ These are nucleated cells mostly involved in providing Immunity
- ↳ WBC's are broadly of two types.

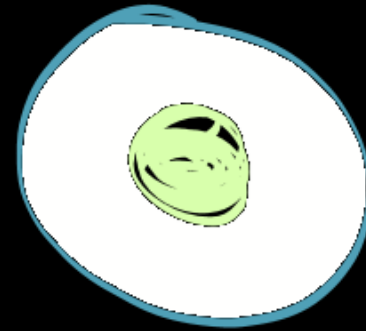
① Granulocyte.



→ Granules (which dye will it take)
- Acidic, Basic, Neutral.

→ Their Cytoplasm contains
Granules (vesicles).

② Agranulocyte



→ Their Cytoplasm lacks this granular appearance

→ Granulocytes Include

- ① Neutrophil (40-60%).
- Most Abundant.
 - Involved in Phagocytosis.



- ② Basophil (<1%)

- Produce:

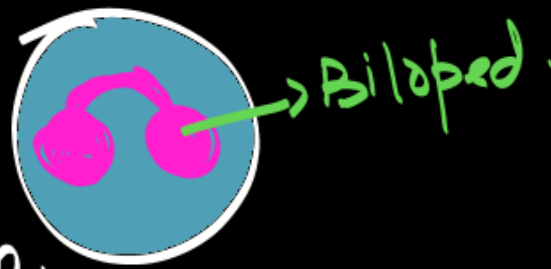
- Histamine (Vasodilator) $\uparrow$ (Inflammation)
- Heparin (Anticoagulant)
- Serotonin (Vasconstrictor) $\downarrow$



- ③ Eosinophil / Acidophil (2-3%)

- Involved in Allergic Reaction.

- Decrease the Allergy (Produce - Antihistamine).

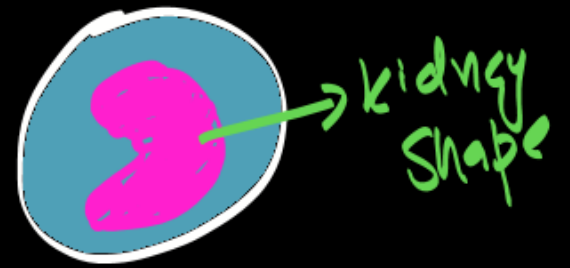


- Agranulocytes Include

- ① Monocytes (6-8%)

Convert in Macrophage in Tissues.

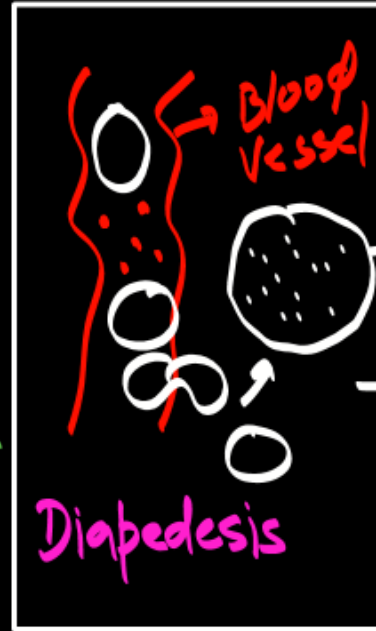
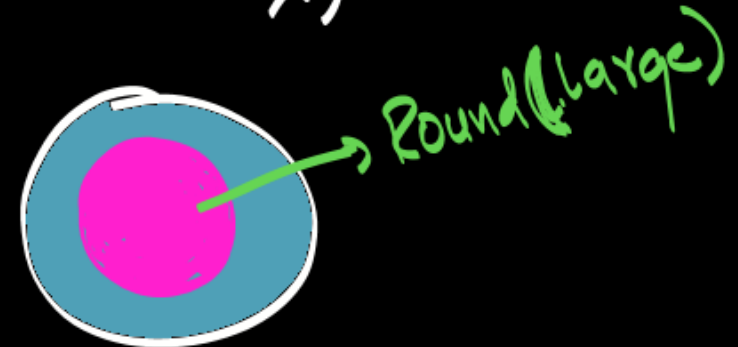
Involved in Phagocytosis.



- ② Lymphocytes (20-25%).

↳ Three types

- B
- T
- Natural killer (NK)

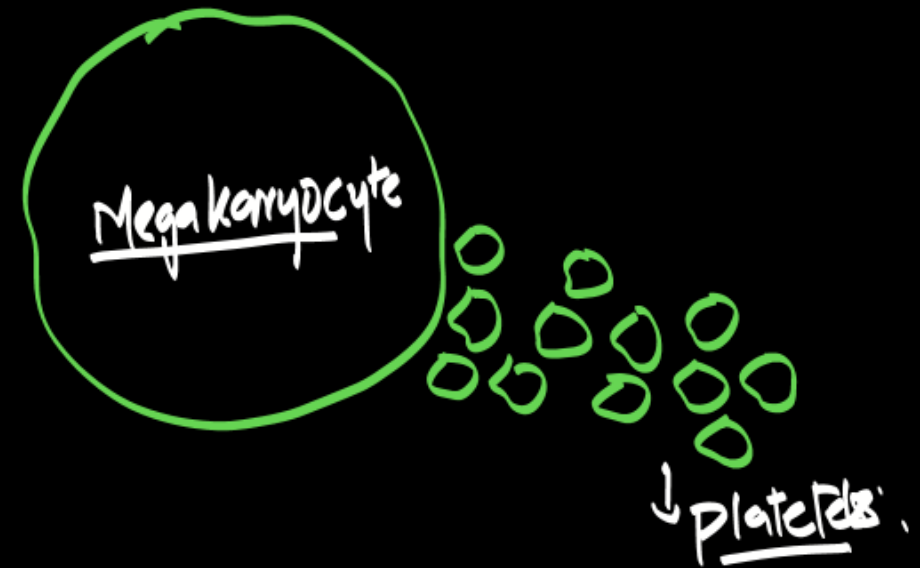


(Phagocytosis)

- Swallowing of Foreign Substances, pathogens etc.

© Thrombocytes (Platelets)

- These are actually cellular fragments from Megakaryocyte cell, in Bone Marrow.
- They are also Anucleated.
- They contain Enzymes for Coagulation.
(Substances)



→ WBC.

→ Number = $6000 - 8000 / \text{mm}^3$.

→ Rise in WBC = Leucocytosis.

→ Decrease in WBC = Leucocytopenia

→ Leukemia - (Blood Cancer)

Type - Rise in WBC Abnormally (Lakhs).

→ Platelets.

Number = $1.5 \text{ lakh} - 3.5 \text{ lakh} / \text{mm}^3$.

→ Rise in platelets = Thrombocytosis.

→ Decrease in platelets = Thrombocytopenia?

Blood Groups.

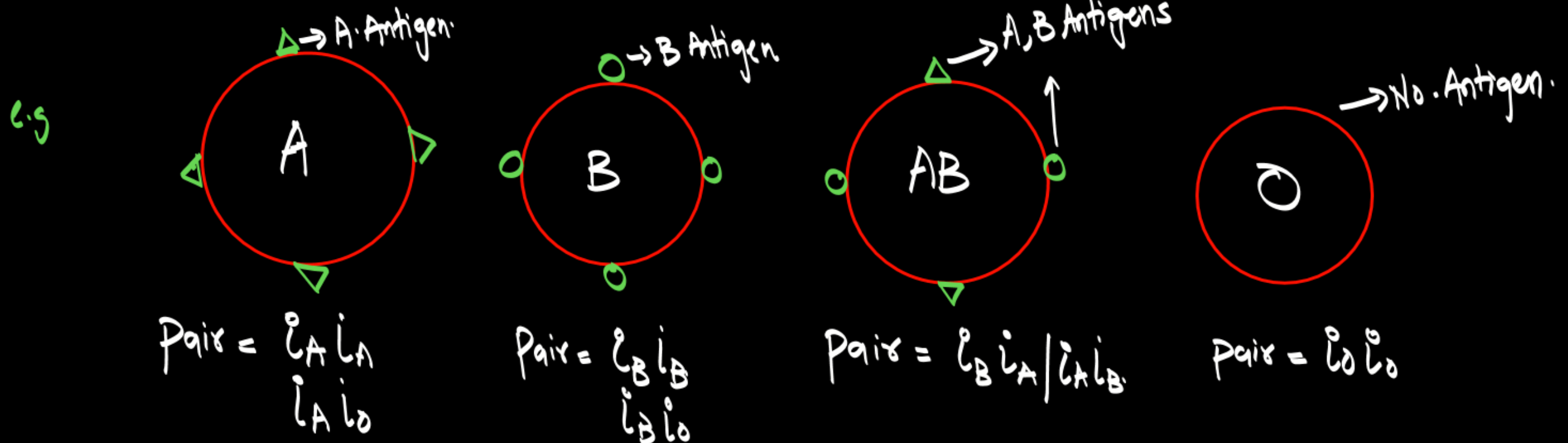
→ ABO type

↳ Most commonly used.

↳ Discovered by Karl Landsteiner in 1900.

↳ According to him, "Blood group is Determined by the Presence of Antigen on RBC Surface."

- Antigen = which causes Antibody Generation
 - usually are Glycoproteins [oligosaccharide]
[oligopeptide]
o-o-o-o-o-o-o-o
- Antigen = Self & Non Self:
- Antibodies are produced against Non-Self.



→ The type of Antigen on RBC Surface is Decided by a pair of Allele.

→ Humans have 3 Alleles for Blood groups but an individual have only pair of it.

→ Types of Alleles are i_A, i_B, i_O . (Multiple Allele)

→ i_A codes for = Antigen A.

→ i_B codes for Antigen B.

→ i_O codes for No Antigen.

→ i_A & i_B are Codominant (Both express)

→ i_A & i_B are Dominant on i_O .

Revision (Class 10th) Mendel.

Allele - Version of Gene.

e.g. T, t

↓ ↓
Dominant Recessive

TT = Tall.

Tt = Tall.

tt = Dwarf.

- Height Decided by pair of Allele.

Possible Genotype.	Type of Antigen	Blood group.	Antibodies
$I_A I_A$	A	A	B
$I_A I_O / I_O I_A$	A	A	B
$I_B I_B$	B	B	A
$I_B I_O / I_O I_B$	B	B	A
$I_A I_B / I_B I_A$	A, B	AB	No Antibody
$I_O I_O$	No Antigen	O	A, B

Question - A person has Antigen A on RBC surface, Antibody Present in His Blood is;

(a) A

(b) B (✓)

(c) A, B

(d) No Antibody.

H/W - If one of the parent has AB Blood group,
Can a child have 'O' Blood group.

Thank
You